

SIMATS ENGINEERING

ASSESSMENT TOOL 2

Name : SHESHATHRI S

Reg no : 1925260316

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to

solve problems effectively.(BL3)

Assessment Tool Weightage: 35%

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

QUESTION NO:

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night)

such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and

constraint checks. State the final valid schedule

1. Hospital Doctor – Shift Assignment — Backtracking Search

Problem Understanding

The hospital has **3 doctors (D1, D2, D3)** and **3 shifts (Morning, Afternoon, Night)**. Each doctor must be assigned exactly one shift, and each shift can contain only one doctor.

The objective is to find a valid assignment satisfying **all five constraints**.

Given Constraints

1. **D1 cannot work Night.**
2. **D2 must work before D3**, meaning $D2 < D3$, according to the order:
Morning < Afternoon < Night.
3. **Only one doctor per shift.**
4. **D3 cannot work Morning.**
5. **Every doctor must receive exactly one shift.** Assessment 2 - Constraint Based ...

Backtracking Search

We assign shifts one doctor at a time and check the constraints after every assignment.

Step 1: Assign D1

Possible values:

D1 = Morning

Check:

- D1 is not Night → ✓
- No other doctor has been assigned → ✓

So:

D1 = Morning

Partial assignment:

Doctor Shift

D1 Morning

D2 —

D3 —

Step 2: Assign D2

Remaining shifts:

Afternoon, Night

Try:

D2 = Afternoon

Check:

- Shift is available → ✓
- D2 must be before D3 → therefore D3 must be after Afternoon.
- D3 cannot be Morning → ✓

So:

D2 = Afternoon

Partial assignment:

Doctor Shift

D1 Morning

D2 Afternoon

D3 —

Step 3: Assign D3

Remaining shift:

Night

Try:

D3 = Night

Check:

- D3 cannot be Morning → Night is allowed ✓
- D2 must be before D3 → Afternoon < Night ✓
- Only one doctor per shift → ✓
- All doctors receive exactly one shift → ✓

Therefore, this assignment satisfies every constraint.

Constraint Check

Constraint	Check	Result
D1 cannot work Night	D1 = Morning	✓
D2 before D3	Afternoon < Night	✓
One doctor per shift	All shifts different	✓
D3 cannot work Morning	D3 = Night	✓

Every doctor gets exactly one shift D1, D2, D3 each assigned ✓

Final Valid Schedule

Shift Doctor

Morning D1

Afternoon D2

Shift Doctor

Night D3

Conclusion

Backtracking Search successfully finds the valid assignment:

D1 → Morning, D2 → Afternoon, D3 → Night.

No backtracking is required after this branch because all constraints are satisfied.

QUESTION NO:

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G).

Obstacles block certain paths.

Constraints:

i. Movement allowed: Up, Down, Left, Right

ii. Cost of each move = 1

iii. Cannot pass through obstacles

iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion

and priority queue updates and determine the optimal path and cost.

2. Robot Grid — BFS Search

Problem Understanding

The robot operates in a **5×5 grid** and must travel from **Start (S)** to **Goal (G)**. It can move **Up, Down, Left, or Right**. Every movement has

a cost of **1**, and obstacles cannot be crossed. The question also specifies the use of **Manhattan Distance heuristic**. Assessment 2-Constraint Based ...

Given Constraints

- Grid size = **5× 5**
- Movement = **Up, Down, Left, Right**
- Cost of each movement = **1**
- Obstacles cannot be entered
- Manhattan distance is specified as the heuristic. Assessment 2-Constraint Based ...

Important Missing Information

The uploaded assessment **does not show the actual 5× 5 grid**, including:

- location of **S**
- location of **G**
- locations of the **obstacles**

Therefore, the exact node-expansion sequence, priority-queue contents, optimal path, and numerical path cost **cannot be determined from the supplied question alone**.

Inventing a grid would make the answer inconsistent with the assessment.

BFS Search Procedure

For the supplied grid, BFS should operate as follows:

1. Put **S** into the queue.
2. Mark S as visited.

3. Remove the first node from the queue.
4. Generate its valid neighbouring cells in the four permitted directions.
5. Reject cells that:
 - o are outside the 5×5 grid,
 - o contain obstacles,
 - o have already been visited.
6. Add valid neighbours to the queue.
7. Continue expanding nodes level by level.
8. When **G** is reached, reconstruct the path using the parent of every node.
9. Since every movement has cost **1**, BFS gives a minimum-cost path in terms of number of moves.

Manhattan Distance

For a current cell (x_1, y_1) and goal (x_2, y_2) :

$$H_n = |x_1 - x_2| + |y_1 - y_2|$$

This represents the number of horizontal and vertical moves needed when obstacles are ignored.

Priority/Frontier Update

The frontier would be updated as nodes are generated:

Initial:

Queue = [S]

After expanding S:

Queue = [valid neighbour 1, valid neighbour 2, ...]

Then the first node is expanded, and newly discovered valid nodes are appended to the queue.

However, the exact queue contents and final path require the missing grid.

So a correct assessment answer should **not fabricate an optimal path or cost** without the grid.

QUESTION NO :

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and

rescue survivors. The robot operates in a partially observable environment where some

paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a

survivor located at G.

Constraints:

- The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed

- The robot has limited sensing capability (can only observe adjacent cells)

- Each movement has a cost of 1, but entering risky zones has an additional cost of 2

·The robot must minimize total cost while ensuring safe navigation

·The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making

process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost

Search / Online Search Agent). Clearly explain the steps involved.

c) Demonstrate how the robot applies AI concepts such as:

·Intelligent agents

·Rationality

·Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the

given constraints.

3. Autonomous Rescue Robot — Constraint-Based Problem Solving

The rescue robot operates in a disaster-affected building represented as a grid. It starts at **S** and must reach a survivor at **G**. The environment is partially observable, some paths are blocked, the robot has limited sensing, and risky zones have additional movement costs. Assessment 2- Constraint Based ...

(a) Analysis of All Constraints

1. Movement Constraint

The robot can move only:

- Up
- Down
- Left
- Right

Therefore, diagonal movement is not allowed.

This reduces the possible actions at each position and makes the search space easier to define.

2. Obstacles

Some cells are blocked and cannot be traversed.

Therefore, before selecting an action, the robot must check whether the next cell is accessible.

An obstacle path must be rejected.

3. Limited Sensing

The robot can observe only **adjacent cells**.

Therefore, it cannot know the complete building layout in advance.

The robot must make decisions using its currently available information.

4. Movement Cost

Each normal movement has:

Cost=1

Therefore, the robot should consider the total cost rather than simply reaching the goal as quickly as possible.

5. Risky-Zone Cost

Entering a risky zone adds an additional cost of **2**.

Therefore:

$$\text{Cost}_{\text{risky movement}} = 1 + 2 = 3$$

A normal movement costs:

$$\text{Cost}_{\text{normal movement}} = 1$$

The robot should therefore consider both distance and risk when selecting a route.

6. Minimum-Cost Requirement

The robot must minimize total cost while maintaining safe navigation.

Thus, a route with fewer movements is not automatically preferable if it contains expensive risky-zone movements.

7. Dynamic Knowledge

The robot must update its knowledge as it moves.

This makes the problem an **online search scenario**.

The robot cannot depend completely on a fixed map because new information can become available during exploration. Assessment 2-Constraint Based ...

(b) Suitable Solution Approach

Uniform Cost Search with Online Search

A suitable approach is to combine **Uniform Cost Search (UCS)** with an **Online Search Agent**.

UCS is appropriate because different actions can have different costs:

- Normal cell $\rightarrow$ cost = **1**
- Risky cell $\rightarrow$ cost = **3**

The robot should expand the available path with the smallest accumulated cost.

Step-by-Step Procedure

Step 1: Start

The robot starts at **S**.

Initial cost:

$g_S=0$

Step 2: Sense the Environment

The robot observes its adjacent cells.

It determines whether neighbouring cells are:

- free,
- blocked,
- risky.

Step 3: Generate Actions

The possible actions are:

- Up
- Down
- Left

- Right

Invalid or blocked movements are removed.

Step 4: Calculate Movement Cost

For a normal cell:

$$g_{\text{new}} = g_{\text{current}} + 1$$

For a risky cell:

$$g_{\text{new}} = g_{\text{current}} + 3$$

Step 5: Select Minimum-Cost Node

The robot maintains a priority queue ordered by the accumulated path cost.

The node with the smallest cost is selected for expansion.

Step 6: Update Knowledge

After moving, the robot senses the newly adjacent cells.

If previously unknown information is discovered, the robot updates its internal map.

Step 7: Replan if Necessary

If the newly discovered information reveals that a path is blocked or more costly than expected, the robot updates the search and selects another available minimum-cost route.

Step 8: Reach G

The process continues until the robot reaches the survivor at **G**.

The selected solution should have the minimum accumulated cost among the paths known to the robot while respecting safety and obstacles.

(c) Application of AI Concepts

1. Intelligent Agent

The rescue robot acts as an **intelligent agent** because it:

Perceives → Processes → Acts → Updates knowledge

For example:

Sensors → Adjacent-cell information → Search/decision → Movement

The robot receives information from the environment and selects an action based on that information.

2. Rationality

A rational agent chooses actions that maximize its performance based on the information currently available.

For this problem, a rational robot should:

- avoid blocked cells,
- consider risky areas,
- minimize total movement cost,
- use available sensor information,
- update its knowledge,
- replan when new information changes the situation.

Thus, rationality means selecting the safest minimum -cost action based on the robot's current knowledge.

3. Environment Type

The environment has several important characteristics:

Property	Description
Partially observable	Robot can observe only adjacent cells
Dynamic/uncertain information	New information can be discovered during movement
Sequential	Current actions affect future decisions
Grid-based	Rooms are represented as cells
Cost-based	Normal and risky movements have different costs
Constrained	Obstacles restrict available actions

The most important characteristic is **partial observability**, because the robot does not initially know the complete environment. Assessment 2- Constraint Based ...

(d) Justification of the Approach

Uniform Cost Search combined with an Online Search Agent is appropriate because the problem has unequal movement costs and incomplete environmental information.

UCS considers the accumulated path cost rather than simply the number of steps. This is important because a normal movement costs 1, while entering a risky zone effectively costs 3.

The online component is necessary because the robot has only limited sensing capability. It cannot rely entirely on a complete map from the beginning. Instead, it must observe adjacent cells, update its knowledge, and revise its route when new obstacles or risky areas are discovered.

Therefore, the approach addresses the major requirements of the problem:

Requirement	How the approach handles it
Obstacles	Blocked cells are excluded
Limited sensing	Robot gathers information online
Different costs	UCS compares accumulated costs
Risky zones	Additional cost is included
Dynamic knowledge	Map is updated during movement
Minimum cost	Lowest accumulated cost is preferred
Safe navigation	Blocked and unsuitable routes are avoided

Final Conclusion

The rescue robot should operate as a **rational intelligent agent** using an **online search strategy with Uniform Cost Search**. It continuously senses its surroundings, updates its knowledge, avoids obstacles, accounts for risky-zone costs, and selects a minimum-cost feasible route toward the survivor.